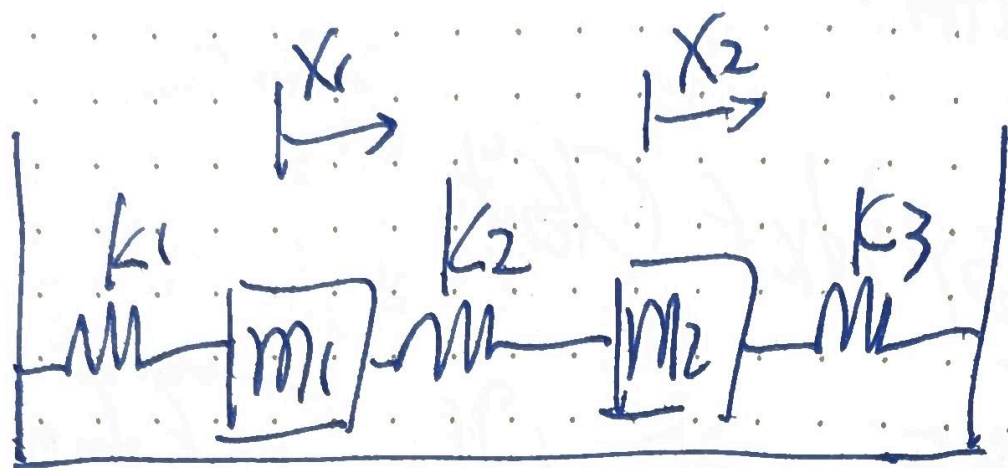


Back to Double Harmonic Oscillators - case B:



no friction!

to solve E.O.M in Lagrangian:

$$\mathcal{L} = T - V \quad \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$

$$T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2$$

$$V = \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_1 - x_2)^2 + \frac{1}{2} k_3 x_2^2$$

$$\mathcal{L} = T - V = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 - \left(\frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_1 - x_2)^2 + \frac{1}{2} k_3 x_2^2 \right)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_1} = m_1 \dot{x}_1$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_1} = m_1 \ddot{x}_1$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = -(k_1 x_1 + k_2 (x_1 - x_2))$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_2} = m_2 \dot{x}_2$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_2} = m_2 \ddot{x}_2$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = -(k_2 (x_1 - x_2) + k_3 x_2)$$

$$m_1 \ddot{x}_1 + (k_1 + k_2) x_1 - k_2 x_2 = 0$$

$$m_2 \ddot{x}_2 - k_2 x_1 + (k_2 + k_3) x_2 = 0$$

or

$$\ddot{x}_1 = -\frac{k_1 + k_2}{m_1} x_1 + \frac{k_2}{m_1} x_2$$

$$\ddot{x}_2 = \frac{k_2}{m_2} x_1 - \frac{k_2 + k_3}{m_2} x_2$$

~~that~~ If def. $V_1 = \dot{x}_1$ $V_2 = \dot{x}_2$ we can get state space matrix of:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{v}_1 \\ \dot{x}_2 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & 0 & \frac{k_2}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & 0 & -\frac{k_2+k_3}{m_2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{bmatrix}$$

we can "shrink" this:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ remove this}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\frac{k_1+k_2}{m_1} & \frac{k_2}{m_1} \\ \frac{k_2}{m_2} & -\frac{k_2+k_3}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

A special "A" matrix, but A matrix indeed

From Taylor chapter 11.2: "A" matrix has the form of:

$$\begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} \frac{1}{m_1} & 0 \\ 0 & \frac{1}{m_2} \end{bmatrix} \begin{bmatrix} -(k_1+k_2) & k_2 \\ k_2 & -(k_2+k_3) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

M^{-1} $-K$

$$\underline{a} = \underline{M}^T \underline{K} \underline{x}$$

$$\underline{M} \underline{a} = \underline{M} \underline{M}^T \underline{K} \underline{x}$$

$$\underline{M} \underline{a} = \underline{K} \underline{x} \leftarrow \text{this form}$$

Back to our "A" matrix, to simplify the eigen-problem, call $m_1 = m_2 = 1$, $k_1 = k_2 = k_3 = 1$

$$\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(-2 - \nu)(-2 - \nu) - 1 = 0$$

$$4 + \nu^2 + 4\nu - 1 = 0$$

$$\nu^2 + 4\nu + 3 = 0$$

$$(\nu + 3)(\nu + 1) = 0$$

$$\nu = -3 \quad \nu = -1 \leftarrow \text{eigenvalue}$$

For $\nu = -3$

$$\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} = -3 \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

$$-2t_1 + t_2 = -3t_1 \rightarrow -t_1 = -t_2$$

$$t_1 - 2t_2 = -3t_2 \rightarrow t_1 = -t_2$$

For $\nu = -1 \rightarrow t_1 = t_2$
 $-2t_1 + t_2 = -t_1$

ν is the "1" of ~~but on the~~ "A" matrix

~~5/12 matrix issue~~

$$\sqrt{-\nu} = \omega$$

$$f = \frac{\omega}{2\pi}$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ eiga vector}$$

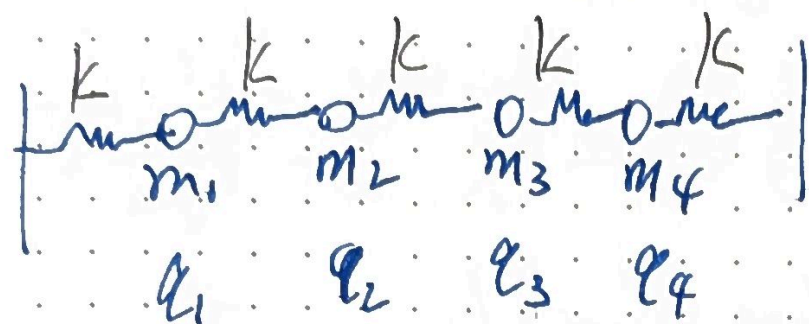
$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ eiga vector}$$

$\Rightarrow D \rightarrow D \rightarrow \infty$

displacement opposite or same

$D \rightarrow D \rightarrow \infty$

Back to chains



$$\mathcal{L} = T - U$$

$$T = \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2 + \frac{1}{2} m_3 \dot{q}_3^2 + \frac{1}{2} m_4 \dot{q}_4^2$$

can be written into matrix form:

$$T = \frac{1}{2} \underline{\dot{q}}^T \begin{bmatrix} m_1 & 0 & 0 & 0 \\ 0 & m_2 & 0 & 0 \\ 0 & 0 & m_3 & 0 \\ 0 & 0 & 0 & m_4 \end{bmatrix} \underline{\dot{q}} = \frac{1}{2} \underline{\dot{q}}^T \underline{M} \underline{\dot{q}}$$

$$U = \frac{1}{2} k (q_1^2 + (q_2 - q_1)^2 + (q_3 - q_2)^2 + (q_4 - q_3)^2 + (-q_4)^2)$$

$$= \frac{1}{2} (k(2q_1^2) + k(2q_2^2) + k(2q_3^2) + k(2q_4^2) - 2kq_1q_2 - 2kq_2q_3 - 2kq_3q_4)$$

can be written into matrix form:

$$U = \frac{1}{2} \underline{q}^T \begin{bmatrix} 2k & -k & 0 & 0 \\ -k & 2k & -k & 0 \\ 0 & -k & 2k & -k \\ 0 & 0 & -k & 2k \end{bmatrix} \underline{q} = \frac{1}{2} \underline{q}^T \underline{K} \underline{q}$$

$$\mathcal{L}(q, \dot{q}) = \frac{1}{2} \underline{\dot{q}}^T \underline{M} \underline{\dot{q}} - \frac{1}{2} \underline{q}^T \underline{K} \underline{q}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0$$

$$\frac{d}{dt} (\nabla_{\dot{q}} \mathcal{L}) - \nabla_q \mathcal{L} = 0$$

Because \$M\$ & \$K\$ matrix is symmetric (as must physics systems are)

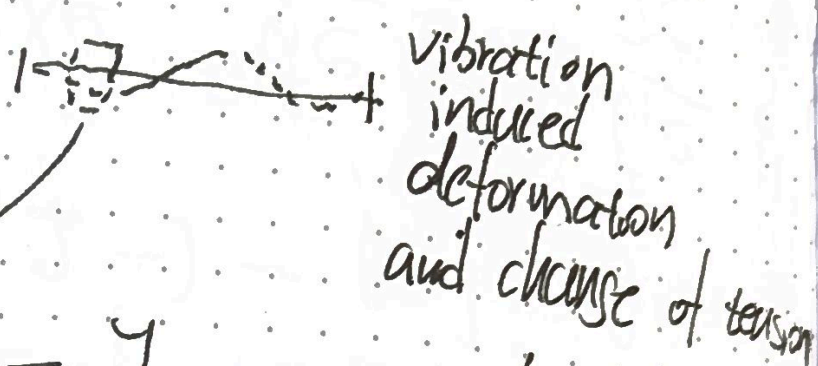
$$\nabla_{\dot{q}} \left(\frac{1}{2} \underline{\dot{q}}^T \underline{M} \underline{\dot{q}} \right) = \underline{M} \underline{\dot{q}} \quad \nabla_q \left(\frac{1}{2} \underline{q}^T \underline{K} \underline{q} \right) = \underline{K} \underline{q}$$

we got $\frac{d}{dt}(\underline{M}\dot{\underline{q}}) - (-\underline{K}\underline{q}) = 0$

that's $\underline{M}\ddot{\underline{q}} + \underline{K}\underline{q} = 0$

eigen-value problem again!

Elastic "string" / media - AKA Field Lagrangian



the $\mathcal{L}$ density = $\mathcal{L}$ per unit length

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

String can only move up and down

$$dT = \frac{1}{2} \rho \left(\frac{dy}{dt}\right)^2 dx$$

$$T = \int_0^t \int_0^L \frac{1}{2} \rho (\dot{y})^2 dx dt$$

$$dV = T (ds - dx) \quad (1)$$

Force $\times$ distance = work

$$V = \int_0^t \int_0^L dV dx dt$$

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Taylor series!

def. $\epsilon = \left(\frac{dy}{dx}\right)^2$, then $ds = f(\epsilon) dx = \sqrt{1 + \epsilon} dx$

$$f(\epsilon) = f(0) + f'(0) \epsilon + \frac{f''(0)}{2!} \epsilon^2 + \dots$$

expanding at $f(0)$

$$f(0) = \sqrt{1 + 0} = 1$$

$$f'(0) = \frac{1}{2} (1 + 0)^{-\frac{1}{2}} = \frac{1}{2}$$

$$f''(0) = -\frac{1}{4} (1 + 0)^{-\frac{3}{2}} = -\frac{1}{4}$$

$$f(\epsilon) = 1 + \frac{1}{2}\epsilon - \frac{1}{8}\epsilon^2 + \dots$$

that's

$$ds = \left(1 + \frac{1}{2} \left(\frac{dy}{dx} \right)^2 - \frac{1}{8} \left(\frac{dy}{dx} \right)^4 + \dots \right) dx$$

too small

$$ds \approx \left(1 + \frac{1}{2} \left(\frac{dy}{dx} \right)^2 \right) dx = dx + \frac{1}{2} \left(\frac{dy}{dx} \right)^2 dx$$

$$ds - dx \approx \frac{1}{2} \left(\frac{dy}{dx} \right)^2 dx$$

$$(2) \rightarrow (1) : dV = T \frac{1}{2} \left(\frac{dy}{dx} \right)^2 = \frac{1}{2} T \left(\frac{dy}{dx} \right)^2$$

$$\therefore Z_{total} = \int_0^L Z_{density} dx = \int_0^L \frac{1}{2} \rho (\dot{y})^2 dx - \frac{1}{2} T (y'')^2 dx$$

$$= \int_0^L \left(\frac{1}{2} \rho (\dot{y})^2 - \frac{1}{2} T (y'')^2 \right) dx$$

$$\cancel{Z(y, \dot{y}, y'')} \quad Z(y, \dot{y}, y')$$

$$\text{or } Z(y, \frac{\partial y}{\partial t}, \frac{\partial y}{\partial x})$$

we need 1D continuous field theory

$$\cancel{\frac{d}{dt} \frac{\partial Z_d}{\partial \dot{y}} + \frac{\partial Z_d}{\partial y'} - \frac{\partial Z_d}{\partial y} = 0}$$

$$\frac{\partial Z_d}{\partial \dot{y}} = \rho \dot{y} \quad \frac{\partial \frac{\partial Z_d}{\partial t}}{\partial \dot{y}} = \rho \ddot{y}$$

$$\frac{\partial Z_d}{\partial y} = -T y' \quad \frac{\partial \frac{\partial Z_d}{\partial x}}{\partial y} = -T y''$$

$$\frac{\partial Z_d}{\partial y} = 0$$

$$\rho \ddot{y} - T y'' = 0$$

$$\frac{\partial y}{\partial t} = \frac{1}{\rho} \frac{\partial y}{\partial x^2}$$

$$\frac{\partial y}{\partial t^2} - c^2 \frac{\partial^2 y}{\partial x^2} = 0 + f(x, t)$$

forcing
↓

Euler-Lagrange form is the most general form of classical Equation of Motion:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$

$\frac{\partial \mathcal{L}}{\partial \dot{q}_i}$ = the ~~canonical~~ ^{canonical} form'
 / conjugate momentum to q_i /

Hamilton called it $P_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$

Generalized momenta

E-L equation turns into:

$$\frac{dP_i}{dt} = \frac{\partial \mathcal{L}}{\partial q_i}$$

ex:

$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2)$ in polar coordinates

$$P_r = \frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}$$

$$\frac{d}{dt} P_r - \frac{\partial \mathcal{L}}{\partial r} = 0$$

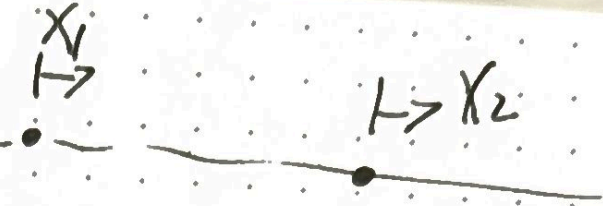
$$m\ddot{r} - m r \dot{\theta}^2 = 0$$

$$\ddot{r} = r \dot{\theta}^2$$

$$P_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$\frac{d}{dt} P_\theta - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\frac{d}{dt} (m r^2 \dot{\theta}) = 0$$

ex: Two particles 

$$L = \frac{m}{2} (\dot{x}_1^2 + \dot{x}_2^2) - V(x_1 - x_2)$$

no cyclic coordinates

potential energy dependent on the distance between two

$$\dot{p}_1 = -V'(x_1 - x_2)$$

$$\dot{p}_2 = V'(x_1 - x_2)$$

$$\dot{p}_1 + \dot{p}_2 = 0 \text{ is conserved.}$$

ex. slightly diff particles:

$$L = \frac{m}{2} (\dot{q}_1^2 + \dot{q}_2^2) - V(aq_1 + bq_2)$$



1910 Emmy Noether

what determines whether there are conservation laws and what are they?